Living Machine Report

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Background

Living machines are systems which use plants to filter contaminated water. In a conventional setup, waste water is sourced and runs through the plants, providing nutrients such as nitrate, phosphoric, and sulfuric compounds which are generally not safe for humans. In a paper on the subject, Dr. John Todd developed a system to treat water from a septic tank using a chain of units: an anaerobic reactor, an anoxic reactor, a clarifier, and ecological fluidized beds.

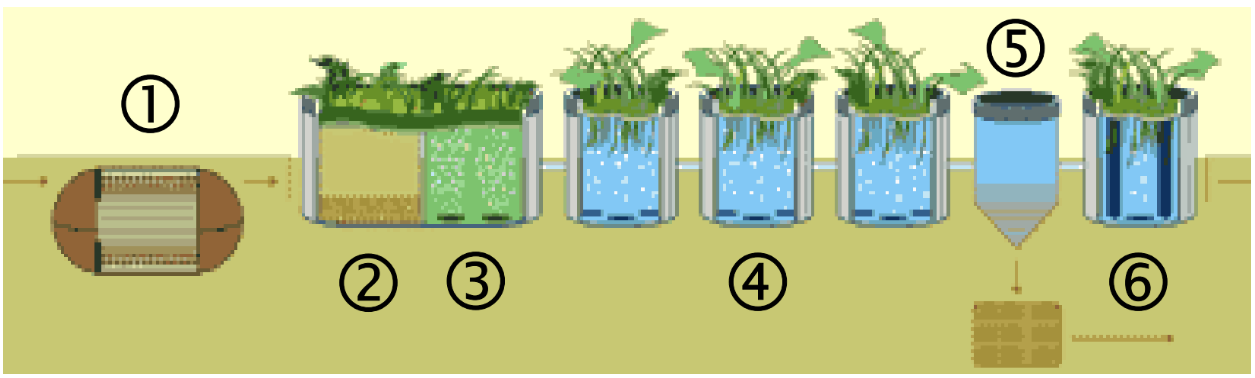


Diagram 1
Living machine pipeline

Anaerobic Reactors:

The primary purpose of the anaerobic reactor is to reduce the concentration of solids in the source water before subsequent filtration steps. These reactors work with gravity as solid material will travel to the bottom of the tank. At the bottom of the tank plastic grids are placed to provide bacteria surface area to help the digestion of solids. The bacteria produce gas which is outgassed through a pipe with carbon meshes implanted to control odors. Sludge is removed from the bottom of the tank and transferred to a biosolids treatment facility, furthering the sustainability factor of this design. At the output of the reactor, 90% of the biological oxygen demand is removed.

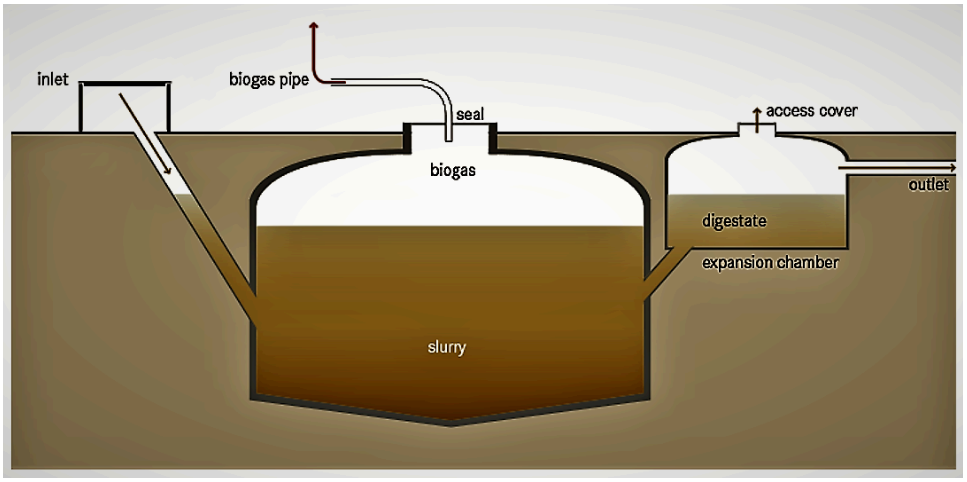


Diagram 2

Anaerobic Reactor

Anoxic Reactors:

The purpose of the anoxic reactor is to encourage the growth of floc-forming microorganisms, which leads to a further decrease of BOD. This is achieved through active mixing of dissolved oxygen and outgasses through an odor control device, usually a biofilter. These reactors can be modified to promote the growth of other bacteria if desired using growth mediums. The residual waste from this reactor is recycled to provide the floc-forming microorganisms with a carbon source.

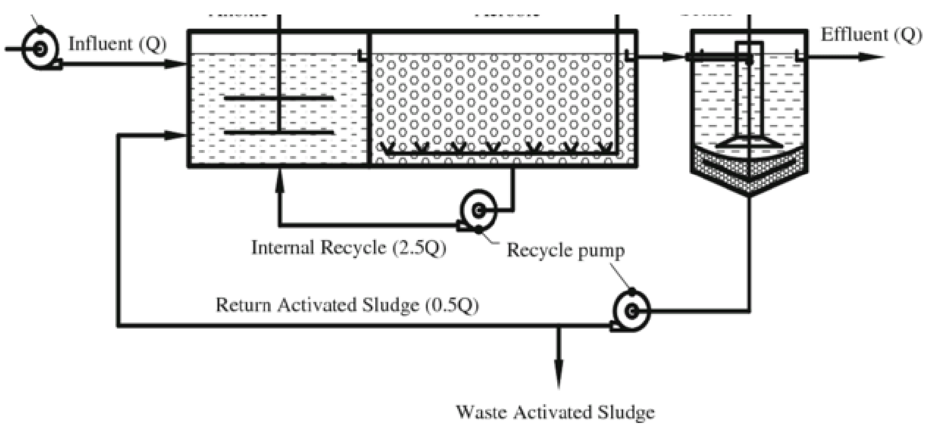


Diagram 3
Anoxic Reactor

Closed Aerobic Reactor

A closed aerobic reactor reduces the BOD further, removes odorous gases, and to stimulate nitrification. Fine bubble diffusers provide aeration to promote offgassing which, similar to the previous reactors, feed into biofilters that sit over the reactor to mitigate odors outside of the reactor. The biofilter is outfitted with vegetation to control moisture levels.

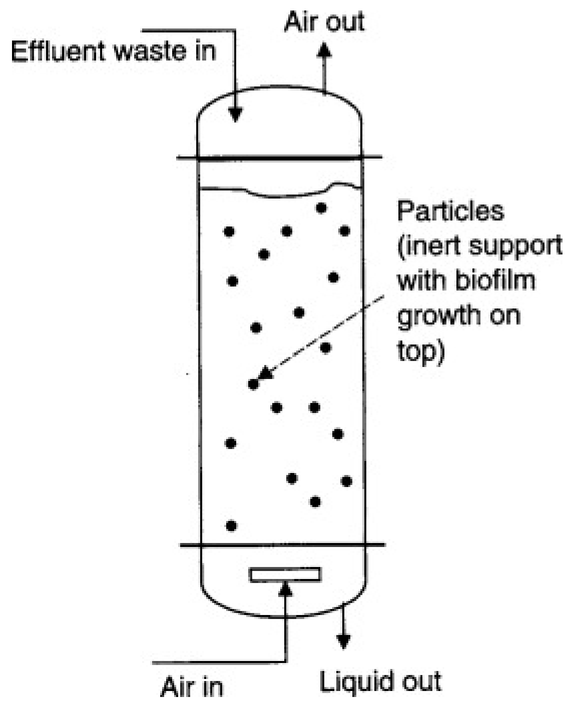


Diagram 4
Closed Aerobic Reactor

Open Aerobic Reactor

These reactors differ from closed aerobic reactors by being covered with vegetation racks which help perform nutrient uptake. These racks also contribute to bacterial growth aswell as serves as an environment for insects and other microorganisms to thrive in.

Clarifier

Clarifiers are tanks where the remaining solids settle to the bottom and are either recycled as carbon sources in previous steps or are discarded in a holding tank for later disposal. On top of the clarifier usually lies duckweed which prevents algae growth by removing nutrients from the water.

Ecological Fluidized Beds

The final step in the process, these perform the final filtering process to achieve an acceptance BOD level. Water runs down between two tanks, one inner and one outer, where growth media and gravel have been placed in the inner tank. The bottom of the inner tank is open to allow filtered waste water to flow into the outer tanks which is pushed back up into the inner tank by air pumps at the bottom of the outer tank. This allows the water to make multiple passes through the filter.

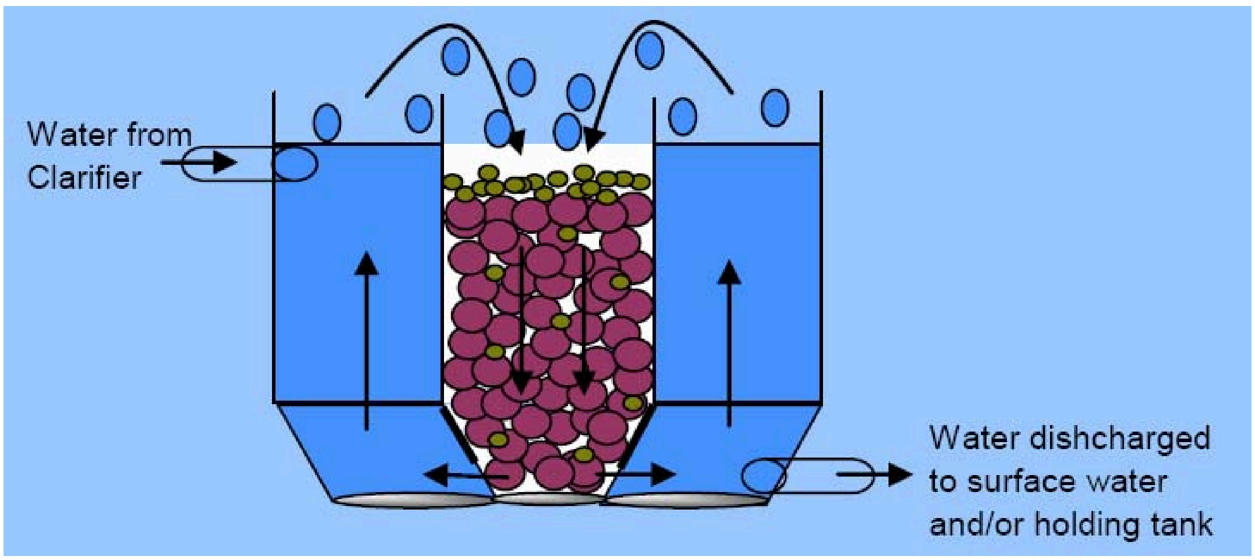
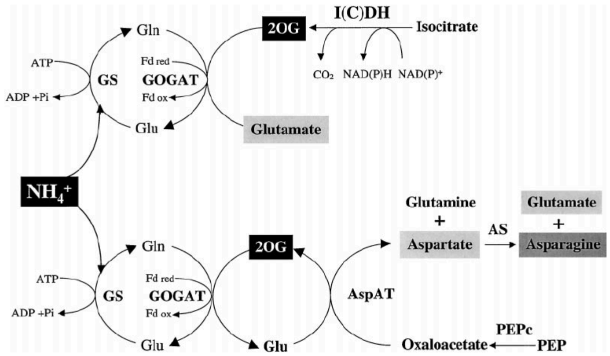
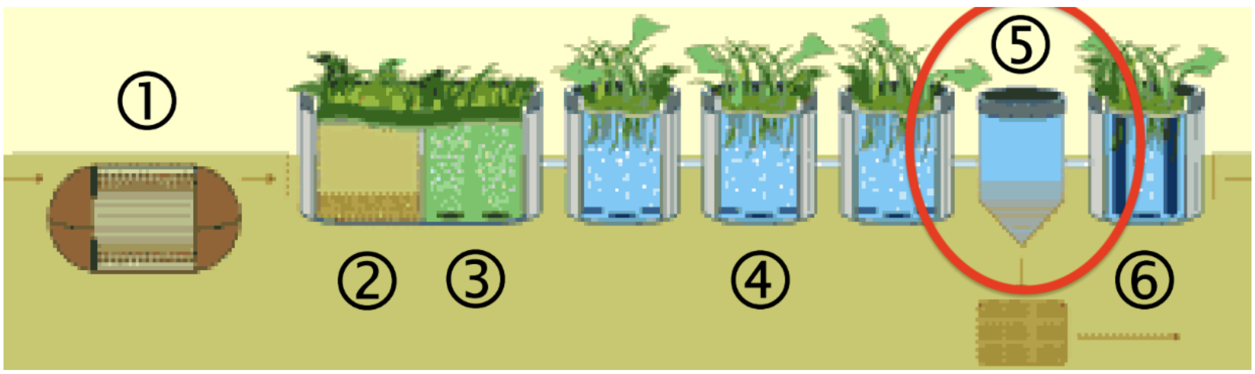


Diagram 5
Ecological Fluidized Beds

Put in a chain, these reactors can provide small to medium operations with limited power resources, making it relatively sustainable because of its semi-closed loop operation. Ethically, there isn't much concern because of the little to no use of unethically sourced materials nor does the system contribute any major source of carbon dioxide compared to other water filtration systems. In this short lab, we looked at the feasibility of using Duckweed and Hornwort to decontaminate water. In regards to the living machine system, our experimental setup follows something similar to the clarifier. Hornwort and Duckweed absorb contaminants such as ammonium directly through the surface of their cells using ammonium transporters. Uncharged ammonia can diffuse directly through the membrane. The nitrogen from the ammonium/ammonia is integrated into amino acids through the Glutamine synthetase-glutamate synthase (GS-GOGAT) cycle.



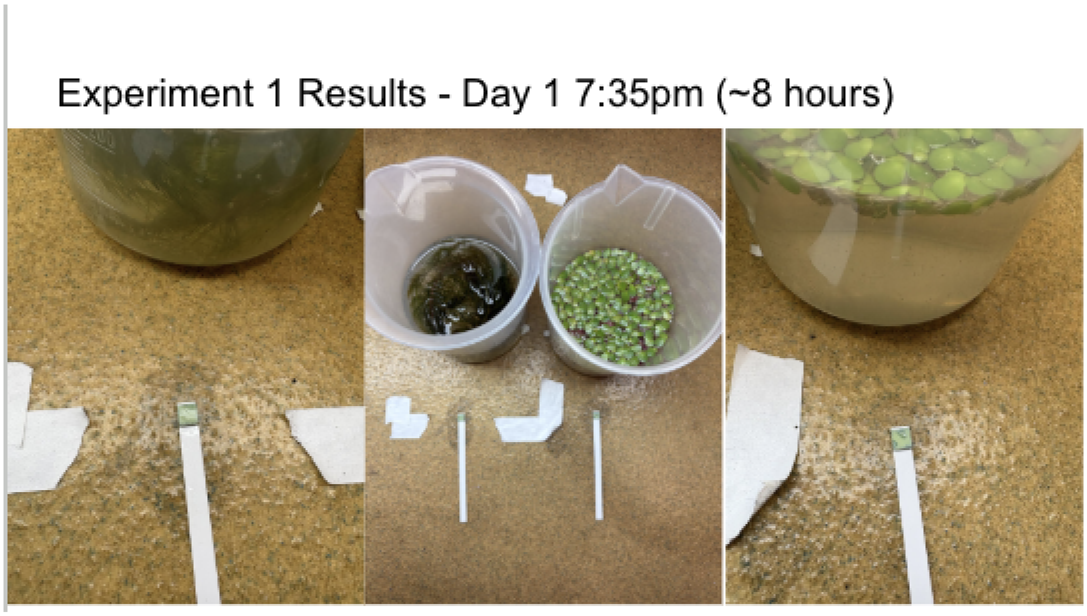
GS-GOGAT Cycle



Systems Diagram

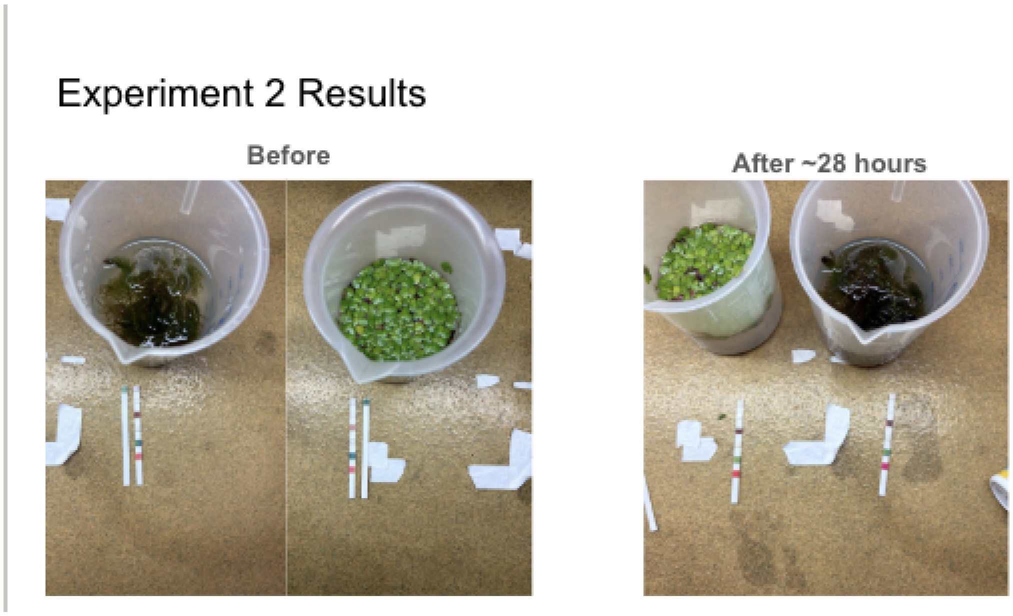
Analysis

Our lab setup consisted of two containers, one containing Duckweed and one containing Hornwort. Each container had 250 mL of water mixed with 10mg of Potassium Sulfate, 10mg of Sodium Phosphate Monobasic, and 10mg of Ammonium Sulfate. Each container was tested for their ammonia levels every 2-3 hours using aquarium test strips. From our experiment, it took roughly 26.3 hours for the ammonia content in the water to reach a safe level which is below 0.5 ppm. The difference between the ammonia uptake between the two plants was negligible, although we did notice that the Hornwort increased the pH of the water from 7.2 to 8.4. We think this is because the Hornwort is submerged more in the water forcing it to take dissolved CO2 from the water.



Experiment 1

Experiment 2



Experiment 3

The benefits of using plants such as Duckweed and Hornworts are immense not only for water filtration, but for general aquatic life. These plants help to remove nitrates which are usually contaminates from farm fertilizers which lead to algae blooms, creating aquatic dead zones because of deoxygenation. The benefits of living machines can be reaped by smaller communities without as many resources which provides more ethical backing to it. The setup requires a few tanks, plumbing, and readily accessible plants that grow extremely quickly and with large volume making this technology suitable for undeveloped nations.

Analysis

After the end of our experiments, the team was curious about testing other types of plants along with the effect of uptake on different contaminates other than nitrate based ones. The reason why we were so narrow in our experimentation was because of setup up time and ease of experimentation. All of the resources were easily available using online shopping and the experimentation didn't require extremely precise or delicate procedures. However, if we had more time, we would take care to take in a wider gamut of plants and test more contaminants, leading to a grid of permutations which would need something more robust than aquatic testing strips. The data collection method would also have to be automated as coming in every few hours to measure the ammonia content was tedious.